

Project Handbook

Rules that apply to every theme

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Contents

AML Project Handbook — rules that apply to every theme	2
1. The component in one table	2
2. Groups	2
3. The GitHub repository — the only thing you submit	2
4. Data on the UPES network	4
5. Milestones	4
6. The 5 group marks — rubric	5
7. The project quiz — 10 marks, written, individual	5
8. Academic integrity	6
9. Getting help	6

AML Project Handbook — rules that apply to every theme

Applied Machine Learning (CSAI2017P) · Autumn 2026 · Dr. Tofik Ali

Read this once. It applies to all eight themes P01–P08. The theme folder tells you *what* to build; this handbook tells you *how it is submitted and marked*.

1. The component in one table

Sub-component	Marks	Granularity	Basis
Project quiz — a written paper, one set per theme	10	per student	your own understanding of your group's work
Repository submission — the state and the history of the repo	5	per group	identical mark for every member

Total 15 of the 100-mark course scale.

The 5 group marks are identical for every member of the group. They are not split by contribution. All individual differentiation happens in the 10-mark written quiz, which every member sits separately. A member who did nothing still receives the group's 5 marks — and then faces a paper about work they did not do. That is the design. Choose your group knowing it.

Both parts are compulsory. A student who does not sit the quiz scores 0 of 10 for it; a group that does not submit a repository link scores 0 of 5.

2. Groups

- **4 to 8 students**, self-formed, from the same section.
- **At most two groups per theme.** Sign-up is first come, first served on the sheet; once a theme has two groups it is closed.
- One member is the **repo owner** — they create the repository and add everyone else as a collaborator. This is an administrative role, not a leadership one, and it earns no extra marks.
- Group membership is frozen at the end of Unit IV. Tell me before that if it changes; after that it does not change.

3. The GitHub repository — the only thing you submit

At the end, you send me **one line: the repository URL**. Nothing else. No zip, no email attachment, no printed report.

3.1 Creating it

- One **public** repository per group, on GitHub, named `aml-2026-<theme-code>-<group-number>` — for example `aml-2026-p03-g2`.
- The repo owner adds **every member as a collaborator**, and adds me as a collaborator too if you choose to keep it private (public is simpler; either is accepted).
- Create it **by the end of Unit IV** — at the start of the work, not at the end of the semester. The creation date is visible and it is part of what I look at.

3.2 What I check when you send the link

I open the repository and read **the commit history**, not just the final files. Specifically:

I look at	What I am checking
<code>git log</code> over time who authored each commit commit messages	that the work was spread across the semester that more than one person actually pushed that they describe changes, not “update”, “final”, “final2”
the two checkpoint tags diffs at each checkpoint	that the checkpoints were real, not backdated that the state then matches what you claimed then

3.3 The commit-history floor

A repository that meets all of these is safe:

1. **Created by the end of Unit IV** and first commit within that week.
2. **At least 20 commits** by the final submission.
3. Commits fall on **at least 6 distinct calendar weeks**.
4. **No more than 40 % of the commits in the final week**.
5. **Every member has commits under their own GitHub account**. Use your own account, set `user.email` correctly, and do not let one person push everyone’s work. This is a condition on the *group’s* repository, so a repository where one person pushed everything loses process marks for the whole group — and the members who never committed will also find the quiz very hard.
6. Two annotated tags, `checkpoint-1` and `checkpoint-2`, pushed **at the time of each checkpoint**.

What fails. A repository whose entire history is 1–3 commits in the last few days before submission — the whole project uploaded at once, files unorganised, one commit called “project” — scores **at most 1.5 of 5** for the group component, however good the code inside it is. This is stated in advance and applied without exception.

Honest exceptions exist. If a genuine problem broke your history — a repo recreated after an accident, a member’s account issue — tell me *at the time*, not at submission. A note in the README written on the day it happened is evidence; an explanation invented at the end is not.

3.4 Repository layout

Every group uses this structure. It is the same for all eight themes; only the number and names of the notebooks vary, and your theme brief tells you which.

```
aml-2026-pNN-gM/  
|-- README.md           # what the project is, how to run it, who is in the group  
|-- requirements.txt     # pinned versions  
|-- .gitignore  
|-- data/  
|   |-- raw/             # the downloaded file(s), unmodified (see §4)  
|   '-- processed/       # generated - usually gitignored  
|-- notebooks/          # numbered in run order; your theme brief names them  
|   |-- 01-eda.ipynb  
|   |-- 02-preprocessing.ipynb  
|   |-- 03-models.ipynb
```

```
|  '-- 04-evaluation.ipynb
|-- src/                # reusable functions imported by the notebooks
|-- results/
|  |-- figures/
|  '-- metrics.csv      # every number you report, in one file
|-- report/
|  '-- report.pdf       # 4-6 pages
'-- REPRODUCIBILITY.md  # §3.6
```

3.5 The README must contain

- The theme code and title, and the group number.
- **Every member: name, enrolment number, GitHub username.**
- One paragraph on the question the project answers.
- The data source URL and the exact download steps.
- How to run it: environment, then the commands, in order.
- The headline result — the metric, the number, and on which split.
- A short “**who did what**” section. It does not change any mark. It exists so that the group agrees, in writing and in advance, on what each person will be able to answer for in the quiz.

3.6 REPRODUCIBILITY.md must contain

- Python version and how to recreate the environment.
- Every random seed used, and where it is set.
- The exact sequence of notebooks/scripts to run.
- Expected runtime and hardware.
- Any manual step (a browser download, a file placed by hand) spelled out.
- Anything that does **not** reproduce exactly, and why.

I will attempt a clean-checkout run on at least a sample of repositories. A project that cannot be re-run is a claim, not a result.

4. Data on the UPES network

The campus TLS appliance breaks Python downloads — `fetch_openml`, `ucimlrepo`, `yfinance`, `requests`, `pip` from some hosts. Details in `../lab-assignments/Anchor-Datasets.pdf`.

So, for every theme:

- **Download the file once in a browser** (at home, on mobile data, or on campus — browsers work), and commit it to `data/raw/` if it is under 50 MB.
- If it is larger than 50 MB, do **not** commit it. Put it in `.gitignore`, and write the download instructions in the README precisely enough that I can reproduce your `data/raw/` myself in under two minutes.
- Mirrors of the raw files are placed on the LMS where licensing permits; check there first.
- **Never disable certificate verification.** `verify=False` or an unverified SSL context anywhere in the repository costs marks. It is the wrong lesson.

5. Milestones

Anchored to units, not dates. Exact dates announced in class.

#	Milestone	Anchored to	Evidence
1	Catalogue released	during Unit IV	—
2	Groups formed, theme selected, repo created	end of Unit IV	repo URL on the sign-up sheet
3	Checkpoint 1 — data loaded, one baseline number	end of Unit V	tag checkpoint-1
4	Checkpoint 2 — full pipeline, initial results	end of Unit VI	tag checkpoint-2
5	Final submission — repo link sent	end of Unit VII	the URL
6	Project quiz — written, individual	final lecture / lab-15 slot	—

Checkpoint 1 is real, not a plan. You show the data actually loaded — shape, head, a chart — and one honest baseline number. Groups whose dataset turns out not to exist are caught here, when there is still time to move. A checkpoint tag pushed after the fact is worse than a missed checkpoint honestly declared.

6. The 5 group marks — rubric

Criterion	Marks	Full marks looks like
Process & commit history	1.0	meets all six floor conditions in §3.3 — including commits from every member’s own account; messages describe changes; both checkpoint tags pushed on time
Completeness	1.5	every item in the theme’s “Done” list is present and demonstrated
Method correctness	1.0	no leakage; split appropriate to the data (stratified / chronological); metric fits the problem and the choice is argued
Reproducibility	1.0	runs from a clean checkout following REPRODUCIBILITY.md; seeds set; reported numbers reappear
Documentation	0.5	README and report are complete, honest about limitations, and readable
	5.0	

The “Further” items in a theme brief are not needed for full marks. They are where a strong group makes the quiz easy for itself.

7. The project quiz — 10 marks, written, individual

- **A written paper. One set per theme**, so groups on different themes sit different papers. Both groups on the same theme sit the same set.
- Conducted in the final lecture / lab slot. Closed-book unless announced otherwise; your own repository may be open on screen.
- Every member sits it **separately**. Answers are your own.
- The questions are **not pre-released**. What is pre-released is the *kind* of question:

1. Trace the path from the raw file to one number in your report.
 2. You reported metric M. Why M and not the obvious alternative?
 3. Here is a block of code from your repository. What does it do? What breaks if line k is removed?
 4. Your result would change if X changed. Up or down, and why?
 5. What is the weakest part of your project, and what would you do with two more weeks?
- **Marking is on understanding, not fluency.** “I don’t know, but I would find out by running Y and looking at Z” earns more than a memorised paragraph that does not fit the question.
 - A contributor answers these comfortably. A passenger cannot, however polished the repository is.

8. Academic integrity

- Using AI tools, tutorials, Kaggle notebooks or Stack Overflow is **allowed and expected** — provided you (a) say so in the README, citing what you used and where, and (b) can explain every line you kept. The quiz tests (b).
- Copying another group’s repository, or submitting a lab assignment as a project, is plagiarism and is reported.
- The eight themes deliberately avoid every lab anchor dataset, so a re-submitted lab is immediately visible.

9. Getting help

Bring a specific question and your repository link. “It doesn’t work” is not a question I can answer; “the chronological split gives an R^2 of -0.3 and I expected it to be positive, here is the cell” is.

Ask early. The gap between Checkpoint 1 and Checkpoint 2 is where projects are won.